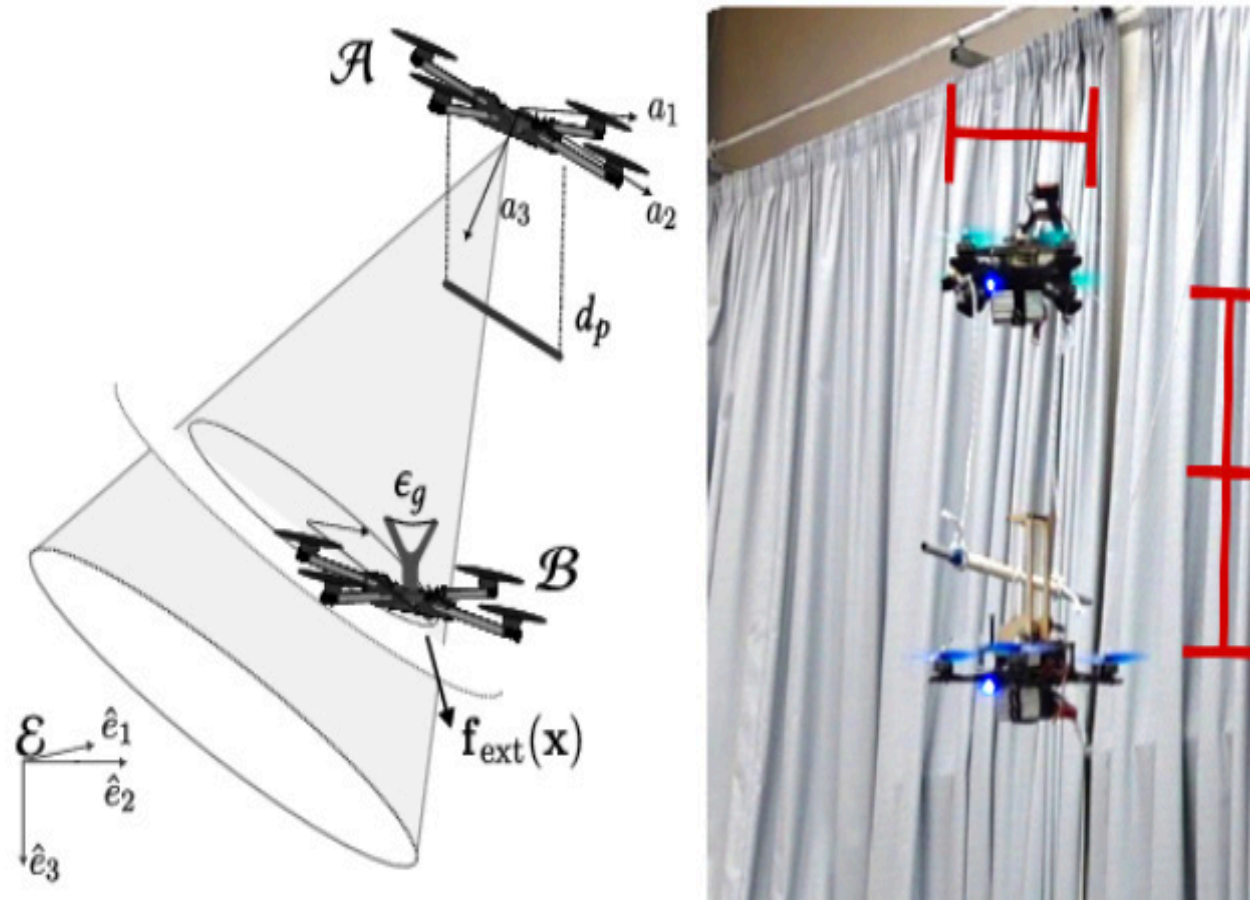


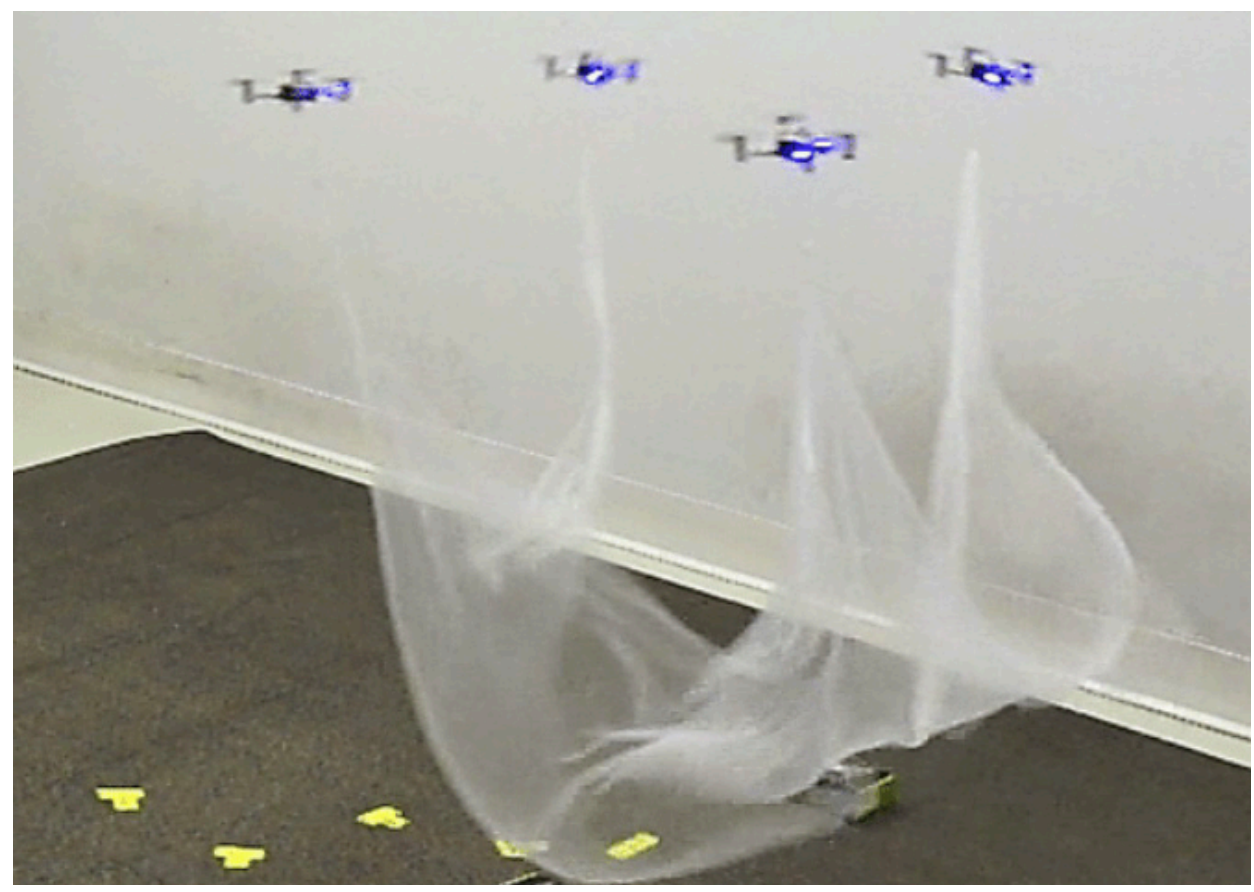
Physics-Based Learning and Control for Downwash in Multi-Quadrotor Systems

Anoop Kiran, Bahar Charyyeva, Maxwell Adorsoo, Zhao Yan
CSCI1470: Deep Learning Final Project, Brown University

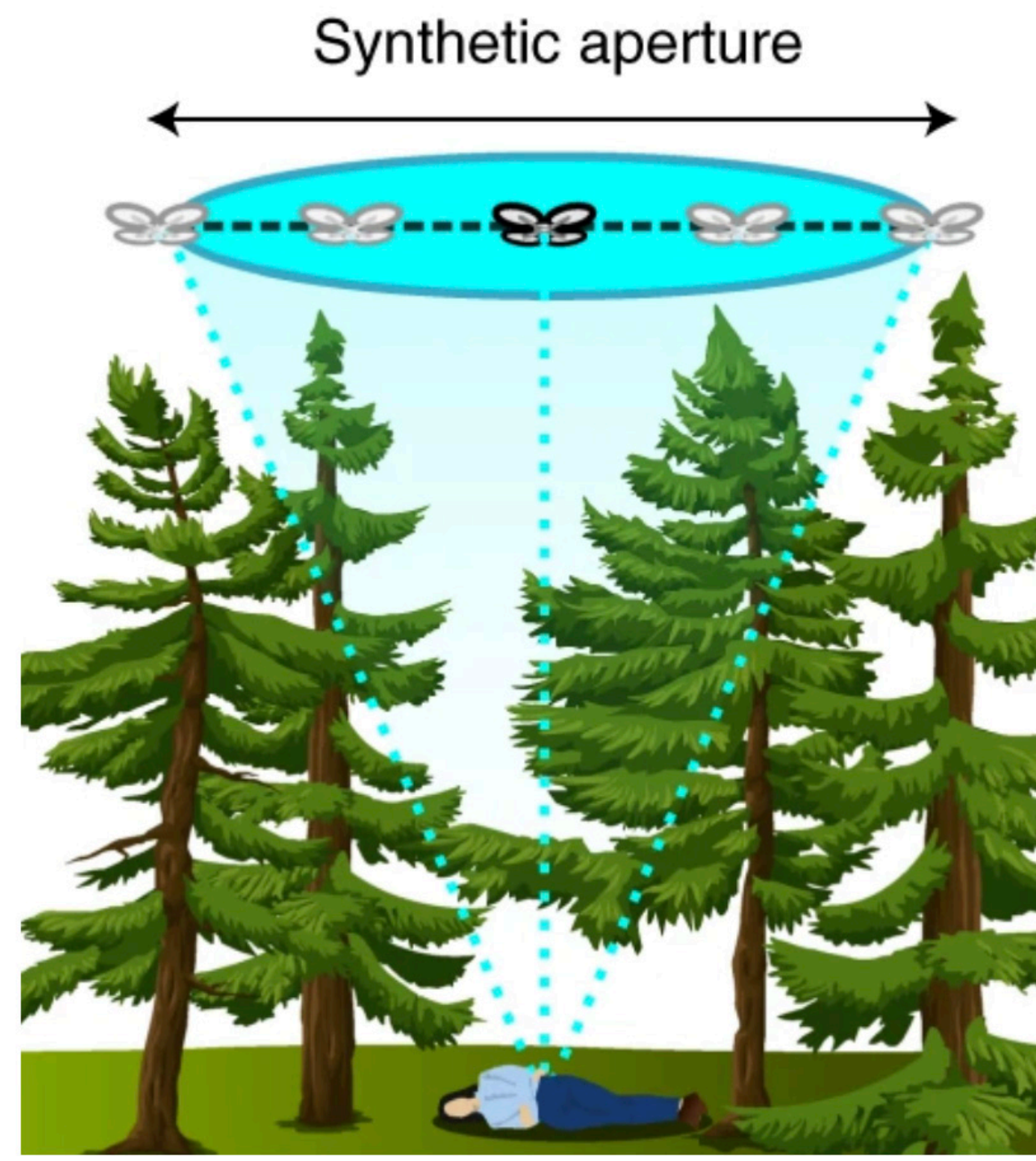
Overview



H.Smith et al., IEEE Trans. on Robotics (2024)



R. Cotsakis et al., IEEE (2019)



D. C. Schedl et al., Science Robotics (2021)

Objective:

Motivation: Stable and controlled flight for safer autonomy in the proximity of quadrotors

Goal: Predict and compensate for aerodynamic disturbances in a two-quadrotor stacked flight configuration using physics-based deep learning approach.

Downwash Effect:

Downwash: Induced airflow generated below quadrotors as propellers rotate

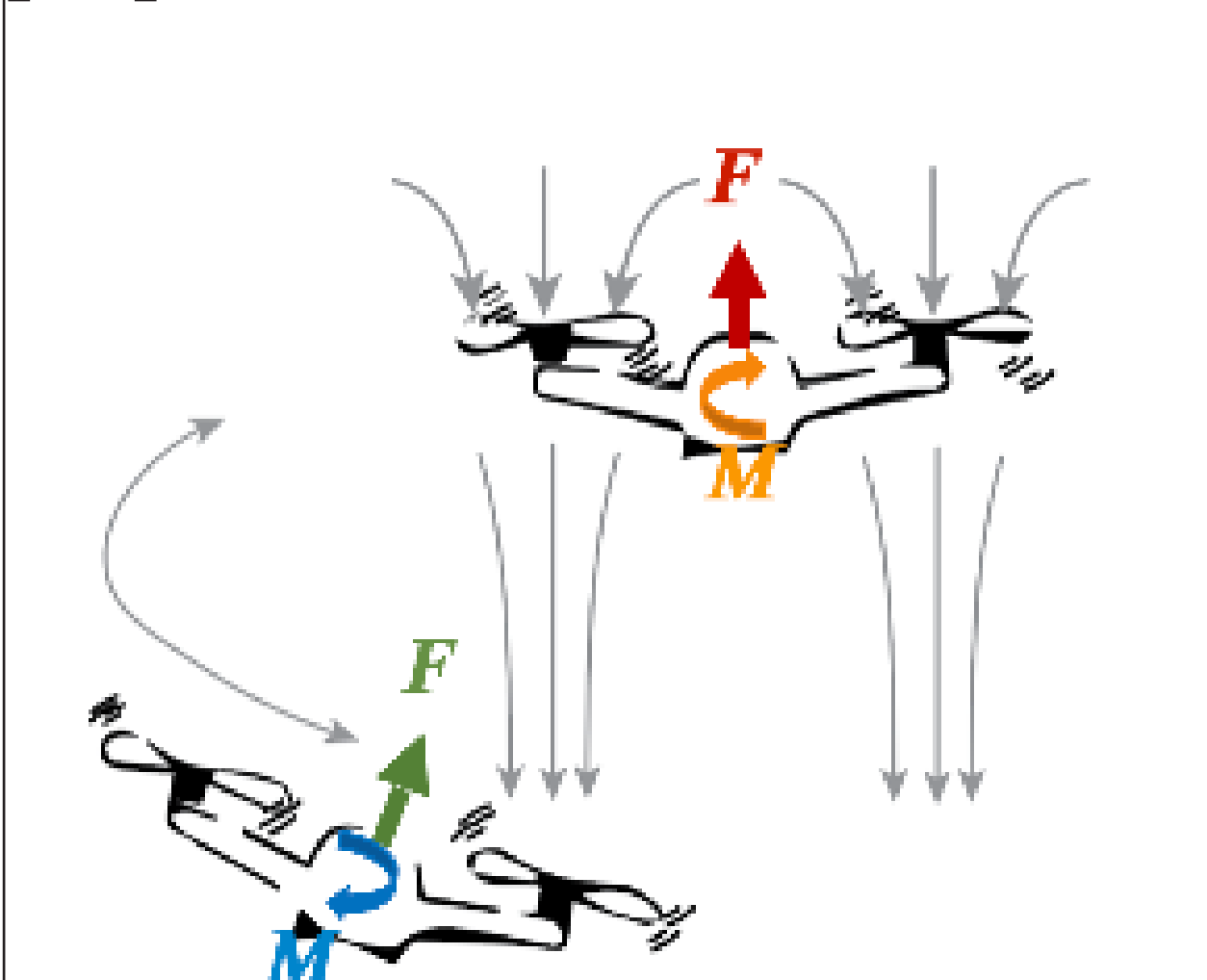


Fig. 1: Schematic of downwash effect

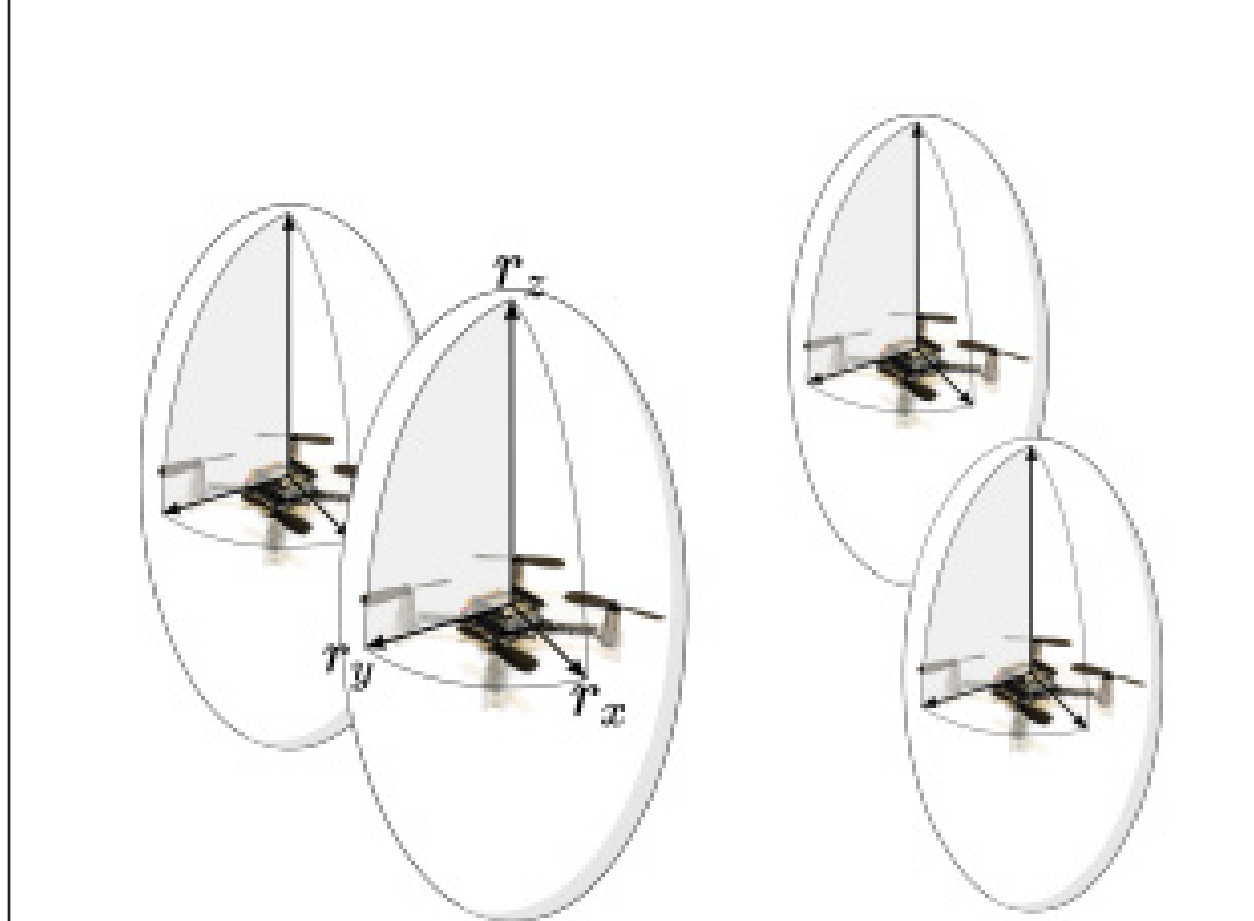


Fig. 2: Collision avoidance workaround

J.A. Preiss et al., IEEE IROS (2017)

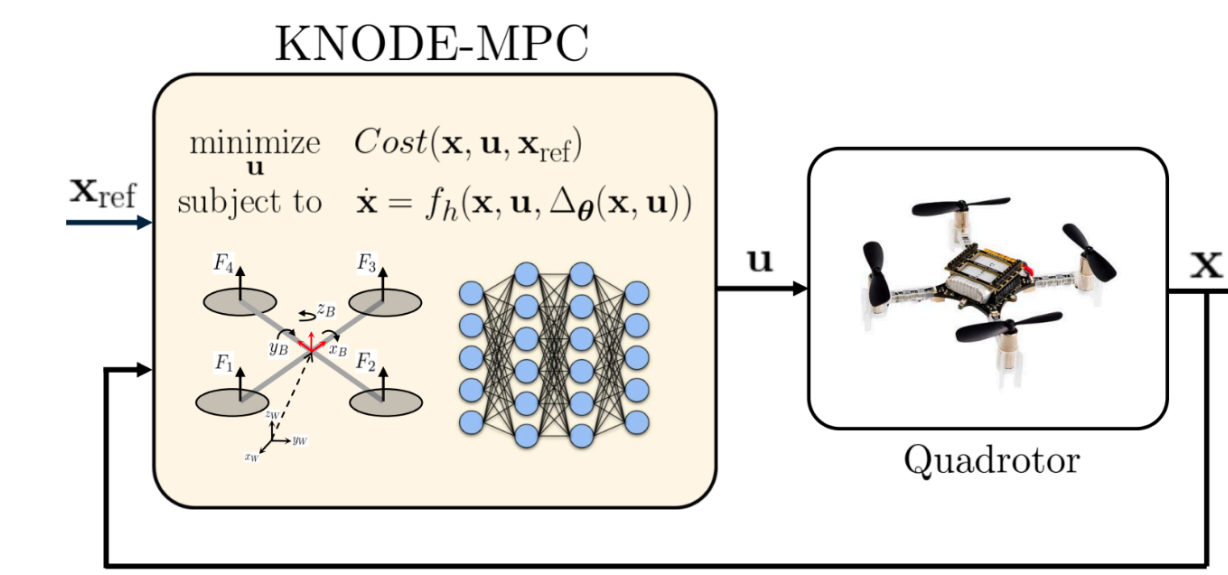
References:

K. Y. Chee, T. Z. Jiahao, and M. A. Hsieh, "KNODE-MPC: A Knowledge-Based Data-Driven Predictive Control Framework for Aerial Robots," IEEE Robotics and Automation Letters, preprint version, accepted December 2021.

Methodology

Baseline:

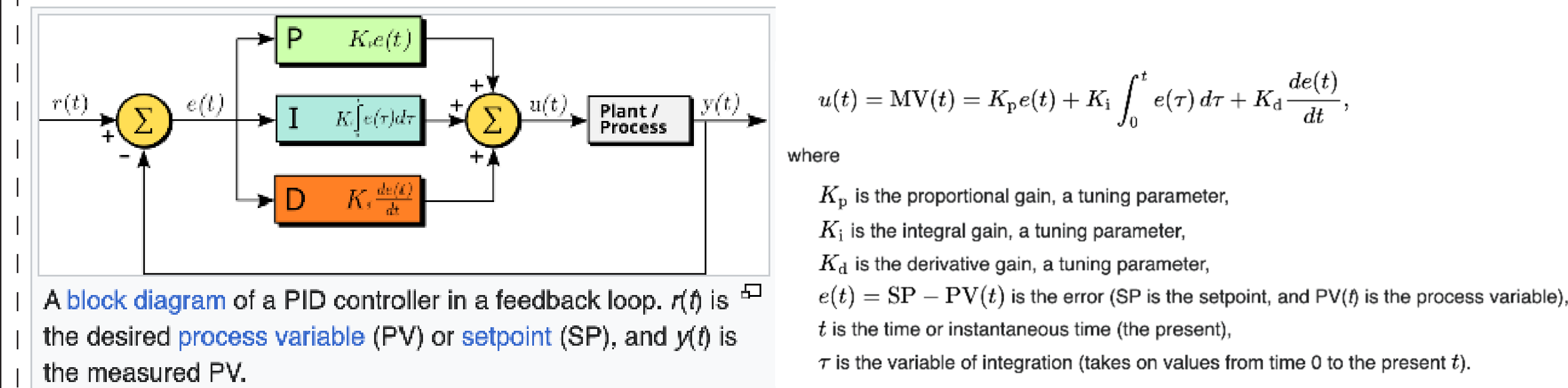
Inspired by prior work on learning residual dynamics for a single quadrotor disturbance force prediction & minimization.



Extension:

- Expand from single to two quadrotors
- Re-implement the model with Proportional Integral Derivative (PID) and physics-based feedforward Neural Network
- Train residual model to capture downwash disturbance effects
- Deeper network & optimal hyperparameters

Control Function:



PID Control. Wikipedia. Accessed: April 28, 2025.

Dataset:

Source: Simulated quadrotor dynamics based on Newton's 2nd Law

Content: State-input pairs (e.g. position, velocity, thrust) sampled from flight trajectories

Model Architecture:

Dimensions: 3-layer MLP with 64, 64, and 32 hidden units.

Input: 12 & **Output:** 3

Activation: ReLU & Linear

Optimizer: Adam

Loss: minimize residual prediction error using Mean Square Error (MSE)

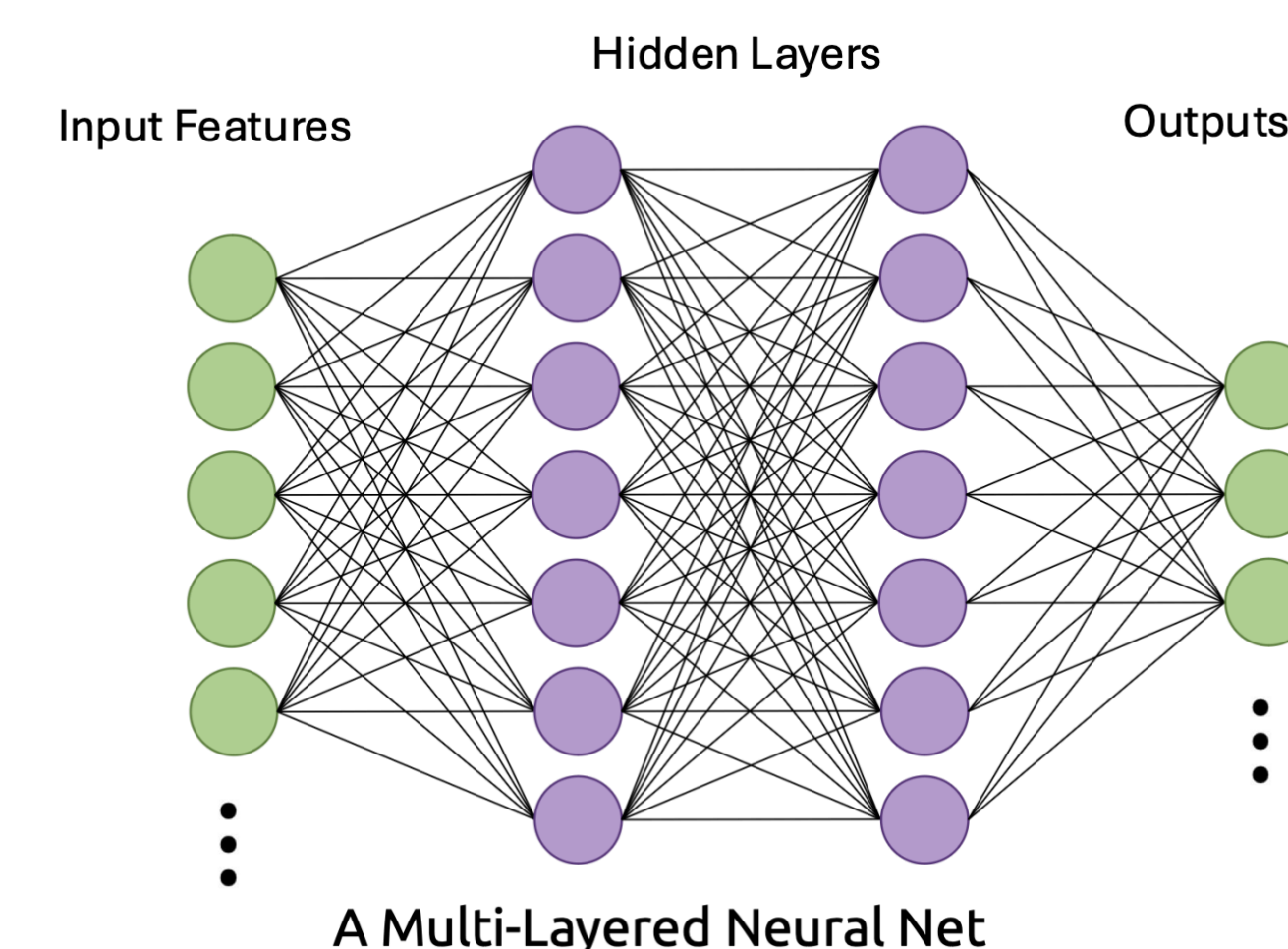
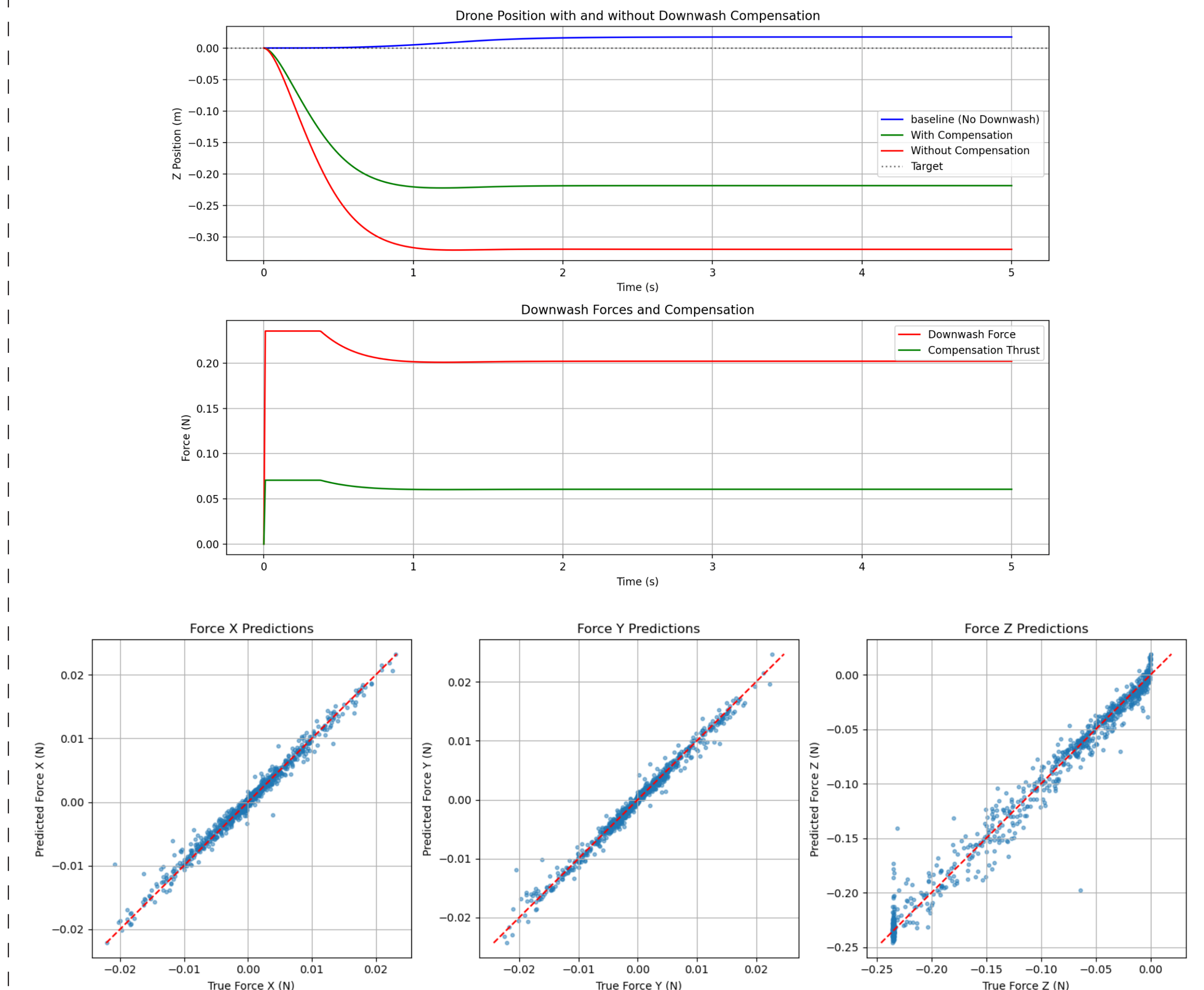


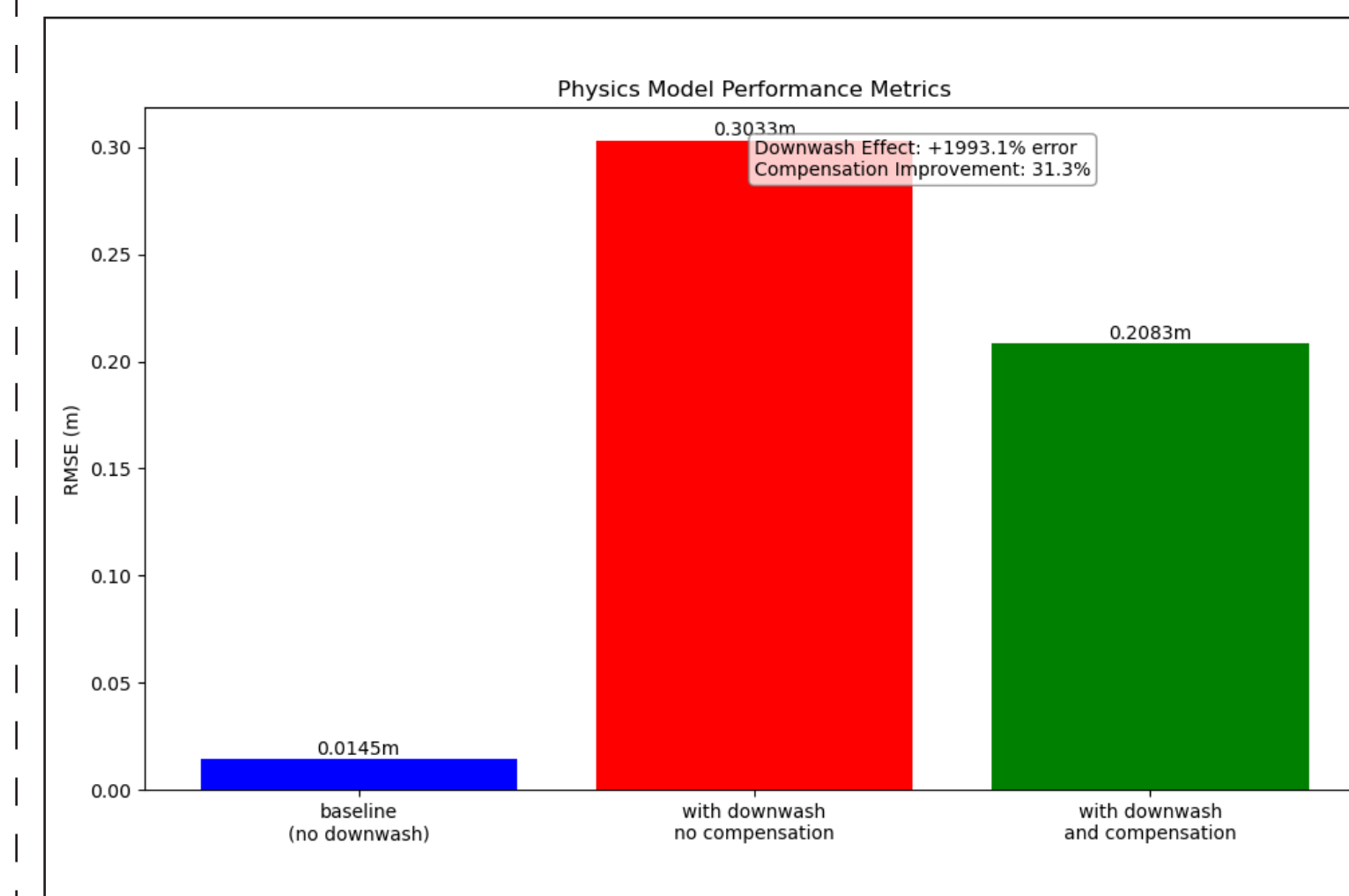
Figure 3: Simple Neural Net Diagram

Results



Kp	Ki	Kd	Comp	Improvement
25	0.5	10	10	30.90%
20	0.7	10	10	31.34%
20	0.5	12	10	30.93%
20	0.5	10	12	30.95%
22	0.6	11	11	31.11%
18	0.4	9	9	30.79%

Figure 4: Improvement across different hyperparameters



Key Findings:

- 31.34% compensation improvement achieved --> 10% higher than original KNODE-MPC paper.
- Deeper network architectures (wider and deeper MLPs) reduce prediction error.

- Tuning PID parameters critically enhances control performance under downwash effects.

Discussion:

- KNODE-MPC offers strong residual dynamic predictions.
- Simpler PID + NN compensation robust for fast development.
- Deeper networks: improved prediction accuracy, optimal PID gains (Figure 4).
- Model achieved 31.34% compensation improvement, exceeding the original paper's 21% improvement by over 10%.
- Incorporating physics-based priors and targeted network scaling can yield strong results even in more complex multi-agent scenarios.